J1A
nuMezzanine_4x34

TCLK 23
T00 32 T00
T01 31 T01
T02 30 T02
T03 29 T03
T04 28 T04
T05 27 T05
T06 26 T06
T07 25 T07
T10 21 T10
T11 20 T11
T12 19 T12
T13 18 T13
T14 17 T14
T15 16 T15
T16 15 T16
T17 14 T17
T20 9 T20
T21 8 T21
T22 7 T22
T23 6 T23
T24 5 T24
T25 4 T25
T26 3 T26
T27 2 T27
TX 12 TX
TY 11 TY

J1C
nuMezzanine_4x34

BCLK 91
B00 100 B00
B01 99 B01
B02 98 B02
B03 97 B03
B04 96 B04
B05 95 B05
B06 94 B06
B07 93 B07
B10 89 B10
B11 88 B11
B12 87 B12
B13 86 B13
B14 85 B14
B15 84 B15
B16 83 B16
B17 82 B17
B20 77 B20
B21 76 B21
B22 75 B22
B23 74 B23
B24 73 B24
B25 72 B25
B26 71 B26
B27 70 B27
BX 80 BX
BY 79 BY

J1E
nuMezzanine_4x34

VCC_TL 1 VCC_TL
VCC_BL 103
VCC_BR 69
VCC_TR 35
3V3_SELECT 34
5V_SELECT 68 GND
GND 33 GND

J1B
nuMezzanine_4x34

LCLK 125 LCLK
L00 134 L0_UNUSED
L01 133 L0_UNUSED
L02 132 L0_UNUSED
L03 131 L0_UNUSED
L04 130 L04
L05 129 L05
L06 128 L06
L07 127 L07
L10 123 L10
L11 122 L11
L12 121 L12
L13 120 L13
L14 119 L14
L15 118 L15
L16 117 L16
L17 116 L17
L20 111 L20
L21 110 L21
L22 109 L22
L23 108 L23
L24 107 L24
L25 106 L25
L26 105 L26
L27 104 L27
LX 114 LX
LY 113 LY

J1D
nuMezzanine_4x34

RCLK 57
R00 66 R00
R01 65 R01
R02 64 R02
R03 63 R03
R04 62 R04
R05 61 R05
R06 60 R06
R07 59 R07
R10 55 R10
R11 54 R11
R12 53 R12
R13 52 R1_UNUSED
R14 51 R1_UNUSED
R15 50 R15
R16 49 R16
R17 48 R17
R20 43 R20
R21 42 R21
R22 41 R22
R23 40 R23
R24 39 R24
R25 38 R25
R26 37 R26
R27 36 R27
RX 46 RX
RY 45 RY

ADDRESS A0–A31

U1A
MC68030RC25B

R27 A2
R26 C4
R20 D13
R21 D12
R22 C13
R24 A4
R24 C12
R23 D11
R25 B13
T01 B12
T02 C11
T00 A13
T06 C10
T04 B11
T02 A12
T03 A12
T07 B10
T05 A11
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
A16 B9
A17 A10
A18 C8
A19 A9
A20 B8
A21 A8
A22 B7
A23 A7
A24 A6
A25 B6
A26 A5
A27 B5
A28 A4
A29 B4
A30 A3
B3 A31
T11 T10 T15 T12 T13 T14 T16 T20 T21 T22 T23 T24 T25 T26 T27

DATA D0–D31

U1B
MC68030RC25B

R07 K13
R06 K12
R04 L13
R02 M13
R03 L12
R05 K11
R00 M12
R01 L11
R27 N13
R25 M11
R24 L10
R26 N12
R22 M10
R23 N11
R20 M9
R21 N10
D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 D14 D15
D16 N9
D17 M8
D18 N8
D19 N7
D20 M7
D21 N6
D22 M6
D23 N5
D24 N4
D25 N3
D26 M4
D27 N2
D28 M3
D29 L4
D30 N1
D31 B17 B16 B15 B14 B13 B12 B11 B10 B07 B06 B03 B05 B02 B01 B04 B00

THREE-STATE BUS CONTROL

U1C
MC68030RC25B

L21 J2
L22 K1
L20 C2
L26 M1
L23 K2
L06 D2
L05 C1
L07 D1
L04 B1
L27 L3
L25 L2
L24 K3
AS CBREQ CIOUT DBEN DS FC0 FC1 FC2 RMC R/W SIZ0 SIZ1

ALWAYS-DRIVEN OUTPUTS

U1D
MC68030RC25B

R17 B2
R10 M2
R15 E13
R16 D3
R11 J13
R12 J12
BG ECS IPEND OCS REFILL STATUS

CYCLE-CRITICAL INPUTS

U1E
MC68030RC25B

L11 F1
L12 G2
L13 G1
L17 L1
L16 J1
L10 E2
L15 H1
L14 H2
DSACK0 DSACK1 STERM CIIN CBACK AVEC BERR HALT

OPEN-DRAIN HOST CONTROL / CLOCK

U1F
MC68030RC25B

TX F12
BY H13
RX G13
LY G12
BX H12
RY F13
TY A1
LX C3
LCLK E1
RESET IPL0 IPL1 IPL2 CDIS MMUDIS BR BGACK CLK

POWER: ALL 10 VCC / 14 GND PINS

U1G
MC68030RC25B

VCC_TL L6
VCC_TL K4
VCC_TL K10
VCC_TL H3
VCC_TL H11
VCC_TL F2
VCC_TL F11
VCC_TL D4
VCC_TL D10
VCC_TL C6
VCC VCC VCC VCC VCC VCC VCC VCC VCC VCC
GND L5
GND L7
GND L8
GND L9
GND J3
GND J11
GND G3
GND G11
GND F3
GND E3
GND E11
GND C5
GND C7
GND C9
GND GND GND GND GND GND GND GND GND GND GND GND GND GND

RESERVED: DO NOT CONNECT

U1H
MC68030RC25B

K5 NC
F4 NC
E10 NC
E12 NC
D5 NC

PWR_FLAG
GND

C1 100nF VCC_TL GND At U1 L6
C2 100nF VCC_TL GND At U1 K4
C3 100nF VCC_TL GND At U1 K10
C4 100nF VCC_TL GND At U1 H3
C5 100nF VCC_TL GND At U1 H11
C6 100nF VCC_TL GND At U1 F2
C7 100nF VCC_TL GND At U1 F11
C8 100nF VCC_TL GND At U1 D4
C9 100nF VCC_TL GND At U1 D10
C10 100nF VCC_TL GND At U1 C6
C11 330pF VCC_TL GND At U1 C6
C12 330pF VCC_TL GND At U1 L6
C13 330pF VCC_TL GND At U1 K4
C14 330pF VCC_TL GND At U1 D4
C15 330pF VCC_TL GND At U1 H3

L0_UNUSED
R1 47k
GND
L0: L00, L01, L02, L03

R1_UNUSED
R2 47k
GND
R1: R13, R14

C16 10uF VCC_TL GND
Bulk: observe polarity

HOST CONTRACT
5V_SELECT grounded: 3V3_SELECT floating. CPU clock: 12.5–25 MHz: timing not yet qualified.
X/Y: open-drain only: keep enabled, release HIGH, never drive HIGH. Internal host pulls used.
Preload all tied unused bank bits LOW before every host-output enable.
Respect byte lanes and bus-release delays; never drive always-output bank.
CPU TTL output-high and host keeper margins are NOT guaranteed. See design-notes.md.

CPGA128 / 5 V / full bus, burst and takeover / experimental direct interface

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Title: Motorola MC68030RC25B DUT daughter board

Size: A2 | Date: 2026-09-13 | **Rev: 1.0**
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